PRACTICE: Mass-to-Mass Stoichiometry **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) Air bags in cars operate according to the reaction below. How many grams of nitrogen gas are produced during the decomposition of 3.25 g Na_3N?

$$2\ Na_3N(s) \rightarrow 6\ Na(s) + N_2(g)$$

1

2) How many grams of lithium are needed to produce 45.0 g of lithium nitride, according to the following process?

$$6\ Li(s) + N_2(g) \rightarrow 2\ Li_3N(s)$$

2

3) A 14.5 g sample of sodium chloride reacts with excess fluorine gas according to the following chemical equation.

$$2\ NaCl(s) + F_2(g) \rightarrow 2\ NaF(s) + Cl_2(g)$$

a. How many grams of sodium fluoride are produced?

b. How many grams of chlorine gas are produced?

a. NaF

3a

b. Cl_2

3b

4) What mass of P_2O_5 can be produced when a 172.1 g sample of phosphorus reacts with an excess of oxygen gas according to the following chemical equation.

$$4\ P + 5\ O_2 \rightarrow 2\ P_2O_5$$

4

5) Determine the mass of lithium hydroxide that is produced when 12.87 g of lithium nitride reacts with an excess of water according to the following process.

$$Li_3N + 3\ H_2O \rightarrow 3\ LiOH + NH_3$$

5

6) Suppose 31.4 g of antimony (III) oxide reacts with excess carbon according to the following process.

$$Sb_2O_3 + 3\ C \rightarrow 2\ Sb + 3\ CO$$

a. What mass of antimony will be produced?

b. What mass of CO will be produced?

c. What mass of carbon is consumed during the reaction?

a. Sb

6a

b. CO

6b

c. carbon

6c

7) A 187.4 g sample of $C_3H_5(NO_3)_3$ decomposes according to the process below.

$$4\ C_3H_5(NO_3)_3 \rightarrow 12\ CO_2 + 10\ H_2O + O_2 + 6\ N_2$$

a. How many grams of carbon dioxide are produced?

b. How many grams of water are produced?

c. How many grams of oxygen gas are produced?

d. How many grams of nitrogen gas are produced?

a. CO_2

7a

b. H_2O

7b

c. O_2

7c

d. N_2

7d